

6. Analysis of Bonds with Embedded Options

1. What are the two drawbacks of the traditional approach to the valuation of bonds with embedded options?

Traditional analysis of the yield premium for a non-Treasury bond involves calculating the difference between the yield to maturity (or yield to call) of the bond in question and the yield to maturity of a comparable-maturity Treasury. The latter is obtained from the Treasury yield curve. The drawbacks of this convention, however, are (i) the yield for all bonds (Treasury versus either callable or noncallable non-Treasury bonds) fails to take into consideration the term structure of interest rates, and (ii) in the case of callable and/or puttable bonds, expected interest rate volatility may alter the cash flow of a bond.

2. What is the static spread for a three-year 9% coupon corporate bond selling at 105.58, given the following theoretical Treasury spot rate values equal to 50, 100, or 120 basis points?

Period	Spot Rate (%)
1	4.0
2	4.2
3	4.9
4	5.4
5	5.7
6	6.0

In traditional yield spread analysis, an investor compares the yield to maturity of a bond with the yield to maturity of a similar maturity on-the-run Treasury security. This means that the yield to maturity of a three-year zero-coupon corporate bond and a 9% coupon three-year corporate coupon bond would both be compared to a benchmark three-year Treasury security. Such a comparison makes little sense because the cash flow characteristics of the two corporate bonds will not be the same as that of the benchmark Treasury.

The proper way to compare non-Treasury bonds (of the same maturity but with different coupon rates) is to compare them with a portfolio of Treasury securities that have the same cash flow. In our problem above, we consider the 9% three-year corporate bond selling for 105.58 (per \$100). The cash flow per \$100 par value for this corporate bond, assuming that interest rates do not change (i.e., assuming static interest rates), is five six-month payments of $(0.09 / 2) \$100 = \4.50 and a payment in three years (6 six-month periods) of $\$100 + \$4.50 = \$104.50$. A portfolio that will replicate this cash flow would include six zero-coupon Treasury securities with maturities coinciding with the amount and timing of the cash flows of the corporate

bond.

The corporate bond's value (105.58 quote per \$100) is equal to the present value of all the cash flows. The corporate bond's value, assuming that the cash flows are riskless, will equal the present value of the replicating portfolio of Treasury securities. In turn, these cash flows are valued at the Treasury spot rates. The period one spot rate is 4%. Dividing by two gives our semiannual discount rate for period one of $4.0\% / 2 = 2.0\%$. Similarly, we can get the remaining five semiannual discount rates for periods two through six which are 2.1%, 2.45%, 2.7%, 2.85%, and 3.0%, respectively. Using our six semiannual discount rates, we get:

$$\begin{aligned} \text{theoretical price} = & \frac{\$4.50}{1.02} + \frac{\$4.50}{(1.021)^2} + \frac{\$4.50}{(1.0245)^3} + \frac{\$4.50}{(1.027)^4} + \frac{\$4.50}{(1.0285)^5} + \frac{\$100 + \$4.50}{(1.03)^6} \\ & = \$4.4118 + \$4.3168 + \$4.1848 + \$4.0451 + \$3.9101 + \$84.5171 = \\ & \$108.3857. \end{aligned}$$

Thus, the price given the Treasury spot rates is \$108.3857. However, the corporate bond's price is \$105.58, which is less than the package of zero-coupon Treasury securities. This is because investors require a yield premium for the risk associated with holding a corporate bond rather than a riskless package of Treasury securities. The **static spread**, also referred to as the **zero-volatility spread**, is a measure of the spread that the investor would realize over the entire Treasury spot rate curve if the bond is held to maturity. It is not a spread off one point on the Treasury yield curve, as is the traditional yield spread. The static spread is calculated as the spread that will make the present value of the cash flows from the corporate bond, when discounted at the Treasury spot rate plus the spread, equal to the corporate bond's price. A trial-and error procedure is required to determine the static spread.

To illustrate how this is done for the corporate bond in our problem we select the first spread of 50 basis points (i.e., 0.5%). To each Treasury (semiannual) spot rate shown in the spot rate column 50 basis points are added. So, for example, the spot rate is $4\% + 0.5\% = 4.5\%$. Dividing by two gives our semiannual discount rate for period one of $4.5\% / 2 = 2.25\%$. Similarly, we can get the remaining five semiannual spot rates for periods two through five which are 2.25%, 2.35%, 2.70%, 2.95%, 3.10%, and 3.25%, respectively. Using our six semiannual spot rates, we get:

$$\begin{aligned} \text{theoretical price} = & \frac{\$4.50}{1.0225} + \frac{\$4.50}{(1.0235)^2} + \frac{\$4.50}{(1.027)^3} + \frac{\$4.50}{(1.0295)^4} + \frac{\$4.50}{(1.031)^5} + \frac{\$100 + \$4.50}{(1.0325)^6} \end{aligned}$$

$$=\$4.4010 + \$4.2957 + \$4.1543 + \$4.0060 + \$3.8630 + \$86.2533 = \$106.9733.$$

Thus, the spot rate plus 50 basis points renders a present value of \$106.9733, which is greater than the corporate bond's price of \$105.58. Thus, the static spread is not 50 basis points but must be a higher spread. We will try 100 basis points.

Proceeding in our trial and error procedure, we add 100 basis points to each Treasury spot rate shown in the spot rate column. So, now the spot rate is 4% + 1% = 5%. Dividing by two gives our semiannual discount rate for period one of 5% / 2 = 2.5%. Similarly, we can get the remaining five semiannual spot rates for periods two through five which are 2.6%, 2.95%, 3.2%, 3.35%, and 3.5%, respectively. Using our six semiannual spot rates, we get:

$$\text{theoretical price} = \frac{\$4.50}{1.025} + \frac{\$4.50}{(1.026)^2} + \frac{\$4.50}{(1.0295)^3} + \frac{\$4.50}{(1.032)^4} + \frac{\$4.50}{(1.0335)^5} + \frac{\$100 + \$4.50}{(1.035)^6}$$

$$=\$4.3902 + \$4.2748 + \$4.1241 + \$3.9673 + \$3.8165 + \$85.0108 = \$105.5838.$$

Thus, the spot rate plus 100 basis points renders a present value of \$105.5838, which rounded off equals that corporate bond's price of \$105.58. **Thus, the static spread appears to be 100 basis points.** Although, we have our answer, we will go ahead and try 120 basis points given in our problem.

So proceeding in our trial and error procedure, we now add 120 basis points to each Treasury spot rate shown in the spot rate column. The spot rate is 4% + 1.2% = 5.2%. Dividing by two gives our semiannual discount rate for period one of 5.2% / 2 = 2.6%. Similarly, we can get the remaining five semiannual spot rates for periods two through five which are 2.7%, 3.05%, 3.3%, 3.45%, and 3.6%, respectively. Using our semiannual spot rates, we get:

$$\text{theoretical price} = \frac{\$4.50}{1.026} + \frac{\$4.50}{(1.027)^2} + \frac{\$4.50}{(1.0305)^3} + \frac{\$4.50}{(1.033)^4} + \frac{\$4.50}{(1.0345)^5} + \frac{\$100 + \$4.50}{(1.036)^6}$$

$$=\$4.3860 + \$4.2665 + \$4.1121 + \$3.9519 + \$3.7980 + \$84.5197 = \$105.0343.$$

As we expected, the spot rate plus 120 basis points renders a present value of \$105.0343, which is less than the corporate bond's price of \$105.58.

3. Why is the investor of a callable bond exposed to reinvestment risk?

Reinvestment risk is the risk caused by reinvesting fixed payments at a lower rate due to an environment of declining interest rates. The investor of a callable bond is exposed to this risk because as interest rates fall there is a greater likelihood that the callable bond will be called by the issuing firm. More details are given below.

The characteristics of a callable bond can work to the disadvantage of an investor because the holder of a callable bond has given the issuer the right to call the issue prior to the expiration date. In brief, callable bonds expose bondholders to loss in value because an issuer will call

a bond at a price below its market value. It will do this when the current yield on bonds in the market is lower than the issue's coupon rate. For example, if the coupon rate on a callable corporate bond is 12% and prevailing market yields are 8%, the issuer may find it economical to call the 12% issue and refund it with an 8% issue. From the investor's perspective, the proceeds received will have to be reinvested at a lower interest rate. This is called **reinvestment risk**. Relatedly, the price appreciation potential for a callable bond in a declining interest-rate environment is limited. This is because the market will increasingly expect the bond to be redeemed at the call price as interest rates fall. This phenomenon for a callable bond is referred to as **price compression**. Because of the disadvantages associated with callable bonds, these instruments often feature a period of call protection, an initial period when bonds may not be called. Also, the investor receives compensation in the form of a higher potential yield.

4. What is negative convexity?

Negative convexity means that the price appreciation will be less than the price depreciation for a large change in yield of a given number of basis points. For a bond that is option-free and displays positive convexity, the price appreciation will be greater than the price depreciation for a large change in yield. It is important to note that a bond can still trade above its call price even if it is highly likely to be called.

5. The theoretical value of a noncallable bond is \$112; the theoretical value of a callable bond is \$108. Determine the theoretical value of the call option.

Effectively, the owner of a callable bond is entering into two separate transactions. First, the owner buys a noncallable bond from the issuer. Second, the owner sells the issuer a call option at a designated option price. The payment received lowers the value of the callable bond. Thus, in terms of price, a callable bond is, generally speaking, equal to the price of the two components parts that we can express as: callable bond price = noncallable bond price – call option price. Rearranging, we have: call option price = noncallable bond price – callable bond price. Inserting our

values, we have:

$$\text{call option price} = \$112 - \$108 = \$4.$$

6. Explain why you agree or disagree with the following statement: “The value of a puttable bond is never smaller than the value of an otherwise comparable option-free bond.”

As described below, one would agree with the statement.

For a puttable bond, the bondholder has the right to sell the bond to the issuer at a designated price and time. A puttable bond can be broken into two separate transactions. First, the investor buys a noncallable bond. Second, the investor buys an option from the issuer that allows the investor to sell the bond to the issuer. The price of a puttable bond is then:

$$\text{puttable bond price} = \text{nonputtable bond price} + \text{put option price}.$$

Given that the value of a put option is never negative and in most (if not all) cases is positive, the value of a puttable bond is always greater than or equal to the value of a nonputtable or noncallable bond or option-free bond. Thus, the statement is correct.

7. What is meant by the option-adjusted spread?

The **option-adjusted spread (OAS)** measures the yield spread (in basis points) that can be used to convert dollar differences between value and price. Thus, basically, the OAS is used to reconcile value with market price. The reason that the resulting spread is referred to as

option-adjusted is because the cash flows of the security whose value we seek are adjusted to reflect the embedded option.